5.5 PHOSPHOLIPIDS

Phospholipids are a characteristic constituent of cellular membranes. A phospholipid is an amphipathic molecule constructed from four components: fatty acids, a platform to which the fatty acids are attached, a phosphate and an alcohol attached to the phosphate. The platform on which phospholipids are built may be glycerol or sphingosine.

Phospholipids derived from glycerol are called phosphoglycerides. The common alcohol moieties of phosphoglycerides are the amino acid serine, ethanolamine, choline, glycerol, and the inositol.

Sphingomyelin is a phospholipid whose backbone sphingosine, an amino alcohol that contains a long, unsaturated hydrocarbon chain.

5.5.1 Types of Phospholipids

There are five general classes of phospholipids based on the type of molecule that is attached to the phosphate group. They are,

1. Phosphatidylserine
2. Phosphatidylcholine
3. Phosphatidylethanolamine
4. Phosphatidylinositol
5. Sphingomyelin

5.5.2 Biosynthesis of phospholipids

Phospholipids can be synthesized by two mechanisms. One utilizes a CDP-activated polar head group for attachment to the phosphate of phosphatidic acid. The other utilizes CDP-activated 1,2-diacylglycerol and an inactivated polar head group. Two fatty acids linked to coenzyme A (CoA) carriers are first joined to glycerol-3-phosphate, yielding phosphatidic acid. A phosphatase then converts phosphatidic acid to diacylglycerol. The attachment of different polar head groups to diacylglycerol then results in formation of phosphatidylcholine, phosphatidylethanolamine, or phosphatidylserine.

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